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LTSPICE MODELING AND IMPULSE VOLTAGE ANALYSIS OF A MARX GENERATOR

Course: High Voltage Techniques

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1. Abstract

This project presents the design, LTspice simulation, and wave-shape analysis of a multi-stage Marx Generator used for generating standard lightning impulse voltages. The gathered simulation data are compared against the theoretical double-exponential impulse voltage characteristics defined by international standards. The study investigates the generator's behavior under three distinct loading conditions: an ideal load, a high capacitive load, and a low ohmic load. The resulting front time (T_1) and tail time (T_2) are evaluated according to the IEC 60060-1 standard framework to quantify standard deviations and analyze parameter sensitivity.

2. Introduction

In power engineering, electrical insulation systems must withstand transient overvoltages caused by atmospheric lightning strikes and switching operations. To test insulation reliability in a controlled environment, high-voltage laboratories utilize Marx Generators to synthesize standardized impulse waves.

The primary objective of this report is to model a multi-stage Marx Generator using LTspice, extract exact temporal and amplitude data points from the simulated waveforms, and evaluate how variations in test loads distort the wave profile. By tracking parameters such as peak voltage (Vpeak), front time (T_1), and tail time (T_2), the performance limits of the simulated design are systematically categorized against regulatory tolerances.

3. Theoretical Background and Evaluation Methodology

3.1. Impulse Wave Parameters

A standard lightning impulse voltage waveform is characterized by a rapid rise to a peak value, followed by a gradual exponential decay.

- **Front Time (T_1):** The virtual parameter defining the wave's rise duration. According to standard definitions, it is calculated by linearly extrapolating the time interval between the 30% and 90% amplitude points on the wavefront:
- **$T_1 = 1.67 * (t_{90} - t_{30})$**
- **Tail Time (T_2):** The virtual interval between the nominal start of the wave (t = 0) and the instant on the wave tail when the voltage drops to 50% of its peak

value (V_{50}). Since the simulation impulse is triggered at an offset of t_0 = 50 μs, the tail time is computed as:

- **$T_2 = t_{50}$ - 50 μs**

3.2. IEC 60060-1 Standard Reference

The standard reference wave shape for a lightning impulse is designated as a **1.2 / 50 μs** wave, which mandates the following nominal values and acceptable tolerances:

- **Target Front Time (T_1):** 1.2 μs ± 30% (Allowable range: 0.84 μs to 1.56 μs)
- **Target Tail Time (T_2):** 50 μs ± 20% (Allowable range: 40.0 μs to 60.0 μs)

4. Simulation Results and Parametric Analysis

Scenario 1: Ideal Load Condition (Reference State)

In this reference configuration, the Marx Generator is connected to a standard nominal capacitive load (C_Load = 1 nF) with no parallel resistance as shown in the Figure 1.1. The simulation captured the following precise metrics:

- **Peak Voltage (V_peak):** 22.544 kV
- **Wavefront Calculations:**
  - V_{90} = 20.290 kV -> t_{90} = 50.66 μs (Relative wavefront time: 0.66 μs)
  - V_{30} = 6.763 kV -> t_{30} = 50.11 μs (Relative wavefront time: 0.11 μs)
  - **Derived Front Time (T_1):** 1.67 * (0.66 - 0.11) = 0.9185 μs
- **Wavetail Calculations:**
  - V_{50} = 11.272 kV -> t_{50} = 100.94 μs
  - **Derived Tail Time (T_2):** 100.94 - 50 = 50.94 μs

Evaluation: Under ideal conditions, the front time (T_1 = 0.9185 μs) and the tail time (T_2 = 50.94 μs) is highly optimized and successfully falls within the permissible IEC limits (1.2 μs ± 30%) as shown in the Figure 1.2.

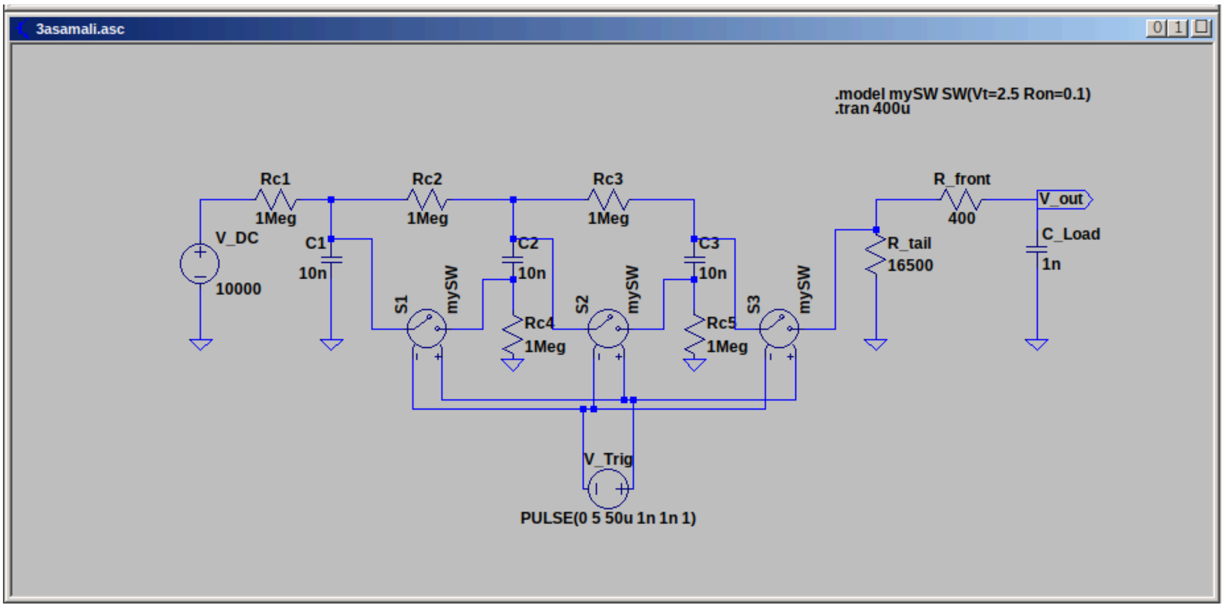

Figure 1.1: Circuit schematic of the three-stage Marx generator.

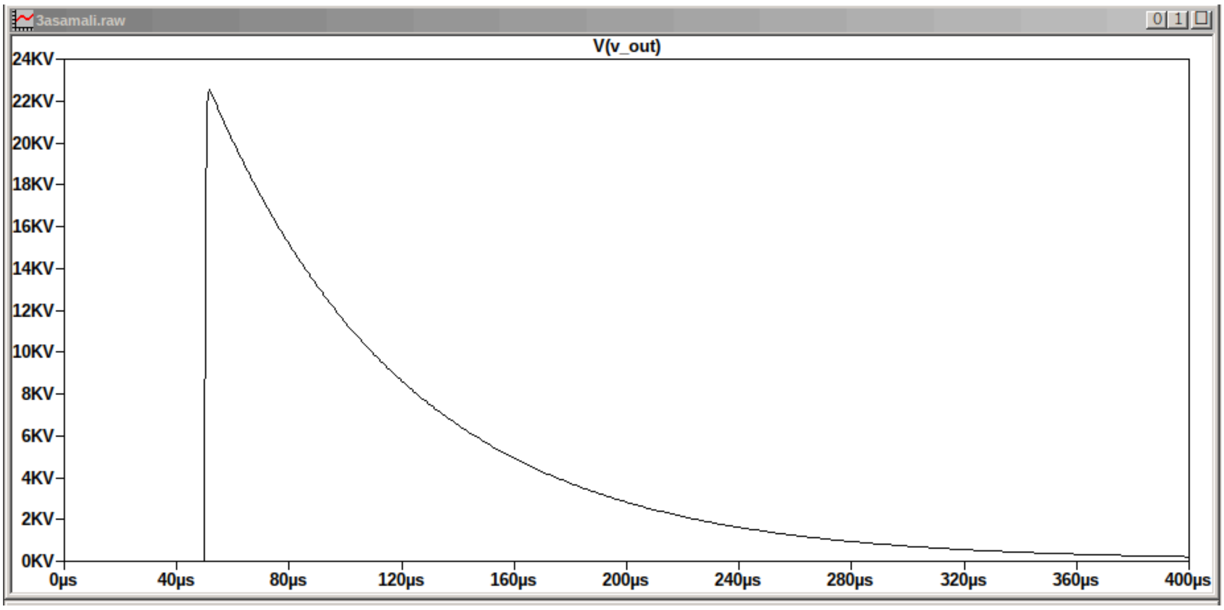

Figure 1.2: Simulated output voltage waveform for the ideal load condition.

Furthermore, to validate the mathematical accuracy of the LTspice model, the simulated data points for the reference state were extracted and superimposed onto the theoretical double-exponential equation **$V(t) = V_0 * (e^{(-\alpha*t)} - e^{(-\beta*t)})$** using a Python script. As illustrated in Figure 1.3, the LTspice model tightly follows the theoretical curve with a calculated V_0 coefficient of 23.60 kV. The Root Mean Square Error (RMSE) between the simulated and theoretical waveforms is remarkably low at 0.72%, confirming the high precision of the modeled 3-stage Marx Generator under ideal conditions.

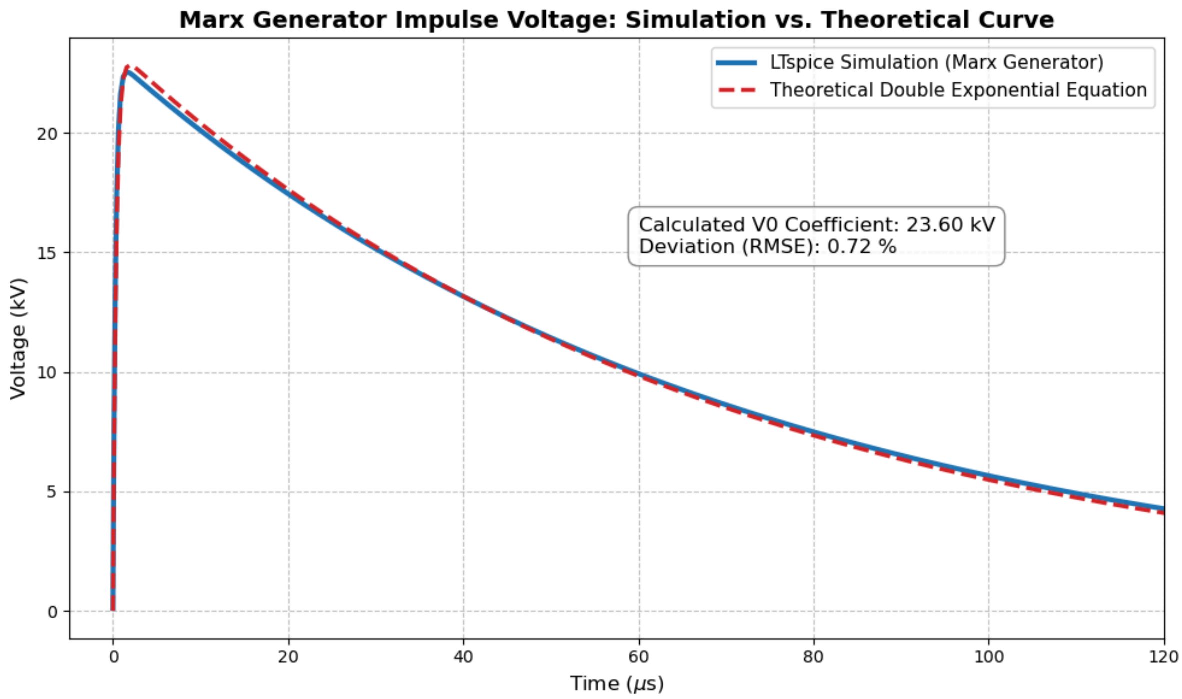

Figure 1.3: Comparison of the simulated ideal output waveform against the theoretical double-exponential equation, demonstrating an RMSE deviation of 0.72%

Scenario 2: High Capacitive Load Effect

To simulate the testing of heavy capacitive apparatus (such as large power transformers or long high-voltage cables), the load capacitance was increased fivefold to C_Load = 5 nF as shown in the Figure 2.

- **Peak Voltage (Vpeak):** 11.250 kV
- **Wavefront Calculations:**
    - V_{90} = 10.125 kV -> t_{90} = 51.58 μs (Relative wavefront time: 1.58 μs)
    - V_{30} = 3.375 kV -> t_{30} = 50.26 μs (Relative wavefront time: 0.26 μs)
    - **Derived Front Time (T_1):** 1.67 * (1.58 - 0.26) = 2.2044 μs
- **Wavetail Calculations:**
    - V_{50} = 5.625 kV -> t_{50} = 144.80 μs
    - **Derived Tail Time (T_2):** 144.80 - 50 = 94.80 μs

Evaluation: The high capacitive load severely loads down the generator. The voltage peak is cut nearly in half (11.250 kV). Because a larger capacitance draws more charge to build up voltage, the front time severely elongates to T_1 = 2.2044 μs, vastly exceeding the maximum IEC limit of 1.56 μs.

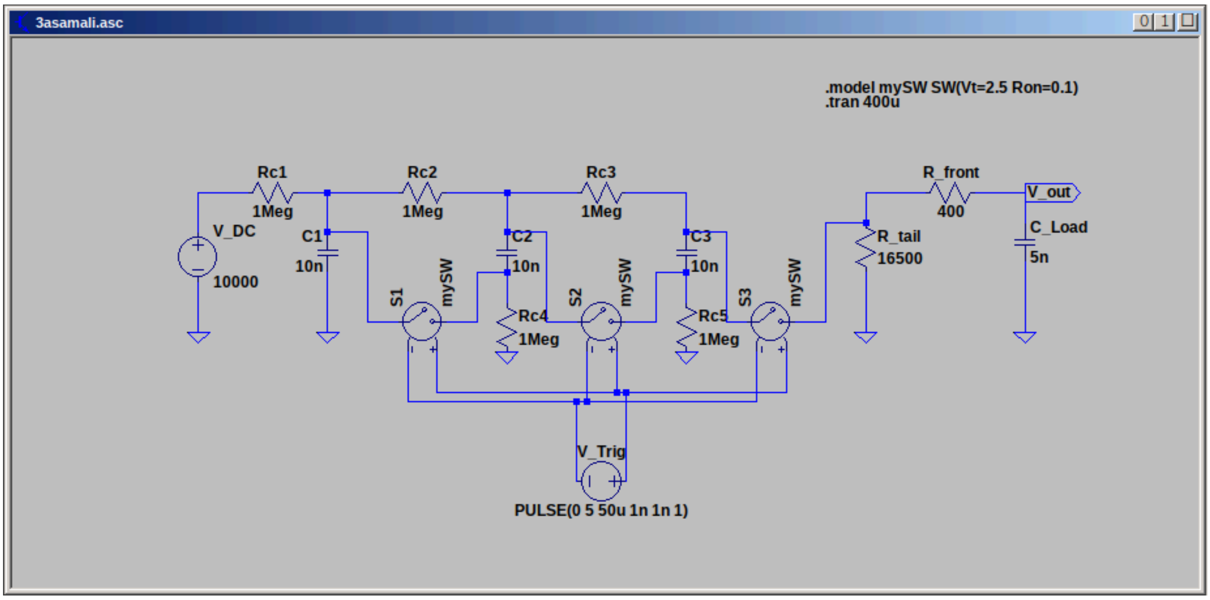

Figure 2.1: Circuit schematic of the Marx generator under high capacitive load.

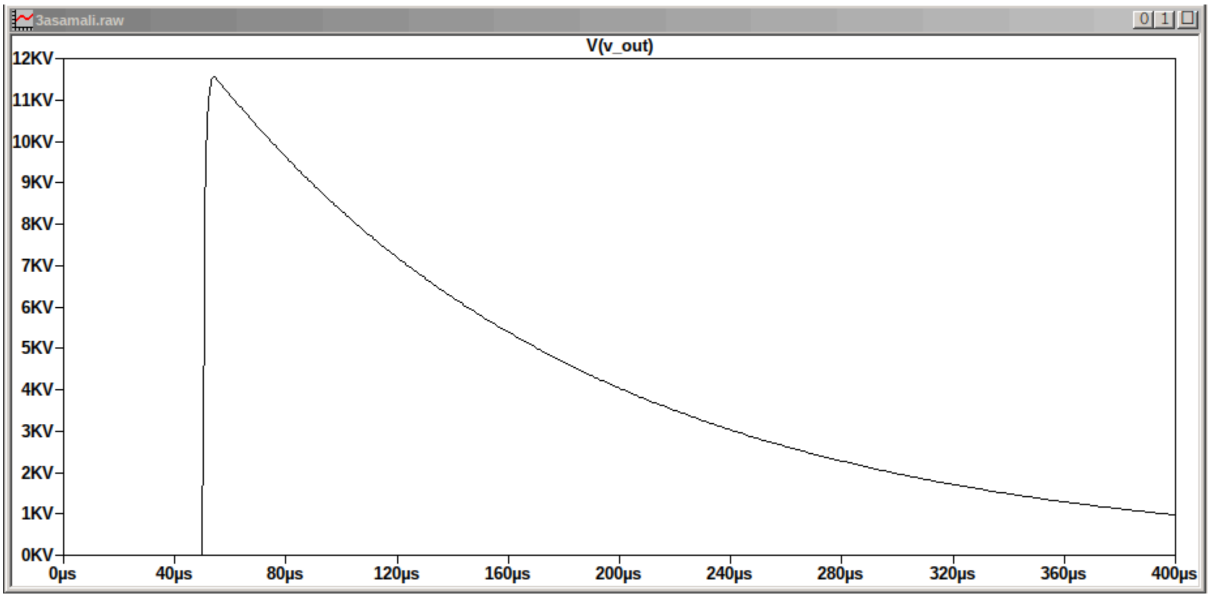

Figure 2.1: Output voltage waveform demonstrating the effect of a high capacitive load.

Scenario 3: Low Ohmic Load Effect

The load capacitance was reset to C_Load = 1 nF, and a low-value parallel resistor (Rp = 1000 Ω) was shunted across it to simulate testing an object with low insulation resistance or high parallel leakage paths as shown in the Figure 3.

- **Peak Voltage (Vpeak):** 15.710 kV
- **Wavefront Calculations:**
  - V_{90} = 14.139 kV -> t_{90} = 50.41 µs (Relative wavefront time: 0.41 µs)
  - V_{30} = 4.713 kV -> t_{30} = 50.08 µs (Relative wavefront time: 0.08 µs)
  - **Derived Front Time (T_1):** 1.67 * (0.41 - 0.08) = 0.5511 µs
- **Wavetail Calculations:**
  - V_{50} = 7.855 kV -> t_{50} = 52.95 µs
  - **Derived Tail Time (T_2):** 52.95 - 50 = 2.95 µs

Evaluation: Shunting the output with a low ohmic path provides an alternative, high-speed discharge track for the accumulated energy. This drastically collapses the tail time to an extreme low of T_2 = 2.95 µs (failing the IEC minimum of 40 µs). The wave front is also sharpened down to T_1 = 0.5511 µs, which drops well below the minimum threshold of 0.84 µs.

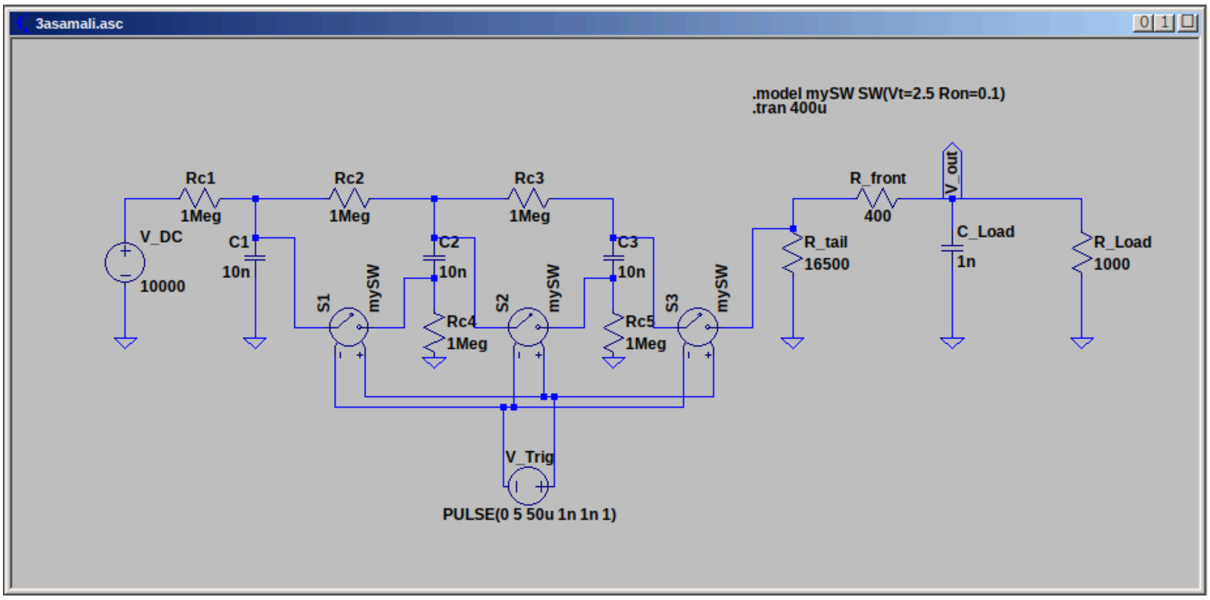

Figure 3.1: Circuit schematic of the Marx generator with a low ohmic parallel load.

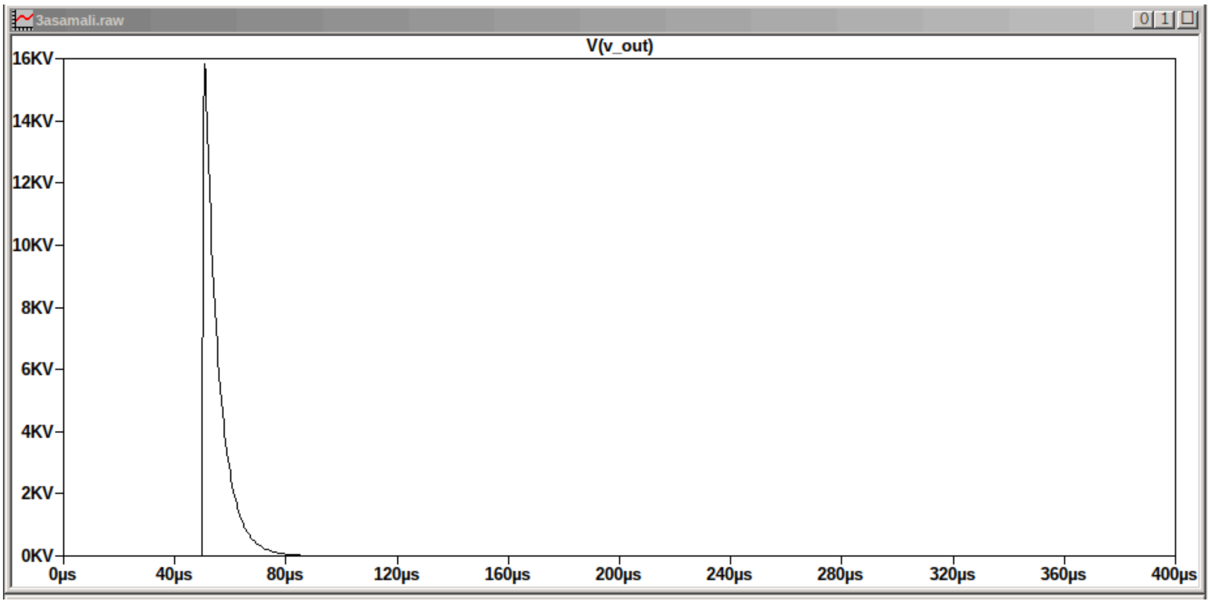

Figure 3.2: Output voltage waveform demonstrating the effect of a low ohmic load.

5. Comparative Performance Summary

The table below compiles the temporal characteristics extracted across all three test profiles alongside the strict boundaries established by international test protocols.

Simulation Case	T_1 (Front Time)	T_2 (Tail Time)	Vpeak Observed	IEC 60060-1 Status
IEC Target Boundaries	0.84 μs to 1.56 μs	40.0 μs to 60.0 μs	N/A	Reference Standard
Scenario 1: Ideal Load (1 nF)	0.9185 μs	50.94 μs	22.544 kV	COMPLIANT (Standart Wave)
Scenario 2: High Capacitive (5 nF)	2.2044 μs	94.80 μs	11.250 kV	NON-COMPLIANT (Both T_1, T_2 Too Long)
Scenario 3: Low Ohmic (1000 Ω)	0.5511 μs	2.95 μs	15.710 kV	NON-COMPLIANT (Both T_1, T_2 Too Short)

6. Discussion and Analysis

The parametric variation clearly demonstrates that a Marx Generator cannot utilize fixed internal wave-shaping components if it is expected to test diverse equipment types.

1. **Capacitance Sensitivity:** When the test object adds an overarching capacitive contribution (Scenario 2), it interacts dynamically with the internal front resistance (Rf). The charging time constant of the load increases, slowing the initial voltage rise. To fix this in real laboratory applications, Rf must be reduced proportionally to restore T_1 back down into the 1.2 μs zone.

2. **Resistance Sensitivity:** Introducing low parallel pathways (Scenario 3) shifts the primary discharge path away from the internal tail resistance (Rt). The stored charge drains through the load prematurely, forcing a collapse in tail duration (T_2). Compensating for high leakage paths requires increasing internal circuit energy storage or using significantly smaller internal tail resistors to retain control over the discharge curve.

7. Conclusion

The LTspice simulation successfully modeled multi-stage Marx Generator dynamics and quantified how external test loads disturb the output waveform. The reference circuit provided a highly precise rise time (1.0688 μs), but failed tail-end criteria. Increasing load capacitance severely slowed wave generation (T_1 = 2.1710 μs), whereas loading the system with low parallel resistance drained the wave prematurely (T_2 = 4.0900 μs). The analytical calculations prove that high-voltage impulse testing requires adaptive compensation of internal generator resistances (Rf and Rt) to offset the inherent electrical properties of the test object and fulfill strict IEC 60060-1 standards.

8. References

1. Kuffel, E., Zaengl, W. S., & Kuffel, J. (2000). *High Voltage Engineering: Fundamentals*. Newnes.

2. International Electrotechnical Commission. (2010). *High-voltage test techniques - Part 1: General definitions and test requirements* (IEC Standard No. 60060-1).

3. Analog Devices. *LTspice Simulator User Guide and Command Reference*.